6-5. An article in *Concrete Research* ("Near Surface Characteristics of Concrete: Intrinsic Permeability," Vol. 41, 1989) presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. The following data are consistent with those reported.

Strength x	Permeability y	Strength x	Permeability y
3.1	33.0	2.4	35.7
4.5	31.0	3.5	31.9
3.4	34.9	1.3	37.3
2.5	35.6	3.0	33.8
2.2	36.1	3.3	32.8
1.2	39.0	3.2	31.6
5.3	30.1	1.8	37.7
4.8	31.2		

EXERCISES FOR SECTION 6-2

(a) Estimate the intercept β_0 and slope β_1 regression coefficients. Write the estimated regression line.

(b) Compute the residuals.

(c) Compute SS_E and estimate the variance.

(d) Find the standard error of ~~the~~ slope and intercept coefficients.

(e) Show that $SS_T = SS_R + SS_E$.

(f) Compute the coefficient of determination, R^2. Comment on the value.

(g) Use a t-test to test for significance of the intercept and slope coefficients at $\alpha = 0.05$. Give the P-values of each and comment on your results.

(h) Construct the ANOVA table and test for significance of regression using the P-value. Comment on your results and their relationship to your results in part (g).

(i) Construct 95% CIs on the intercept and slope. Comment on the relationship of these CIs and your findings in parts (g) and (h).

(j) Perform model adequacy checks. Do you believe the model provides an adequate fit?

(k) Compute the sample correlation coefficient and test for its significance at $\alpha = 0.05$. Give the P-value and comment on your results and their relationship to your results in parts (g) and (h).

6-11. Consider the data and simple linear regression model in Exercise 6-5.

(a) Find the mean permeability given that the strength is 2.1.
(b) Compute a 99% CI on this mean response.
(c) Compute a 99% PI on a future observation when the strength is equal to 2.1.
(d) What do you notice about the relative size of these two intervals? Which is wider and why?